COLLABROOM

PRESENTED BY-

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BRIEF ABOUT THE PROJECT

- This project focuses on developing a full-stack video conferencing web application.
- The system allows users to create, join, and schedule meetings online.
- It provides real-time video and audio communication.
- The application ensures secure authentication and smooth user experience.
- It is designed to simulate modern online meeting platforms.

OBJECTIVE

- **Develop a real-time communication platform:** Build a web application enabling users to create and join.
- **Secure and user-friendly experience:** Provide secure authentication and an intuitive interface for smooth usage.
- **Apply full-stack development concepts:** Integrate modern frontend, backend, and APIs for scalable system development.

KEY FEATURES

- User authentication and authorization
- Create instant meetings
- Schedule future meetings
- Join meetings via link or ID
- Real-time video and audio streaming
- Meeting history and recordings
- Responsive and modern UI

COMPARISON WITH THE EXISTING SYSTEMS

FEATURE	EXISTING SYSTEMS	PROPOSED SYSTEM
Cost	Subscription Based	Free Prototype
Customisation	Limited	Fully customisable
Learning Purpose	Not Applicable	Educational
Scalability	High	Moderate
Features	Advanced	Core Features

TOOLS BEING USED FOR PROJECT

FRONT END
- Next.js
- React
- Tailwind CSS

AUTHENTICATION
- Clerk

DATABASE
- Cloud database services

DEVELOPMENT TOOLS
- VS Code
- GitHub
- Postman

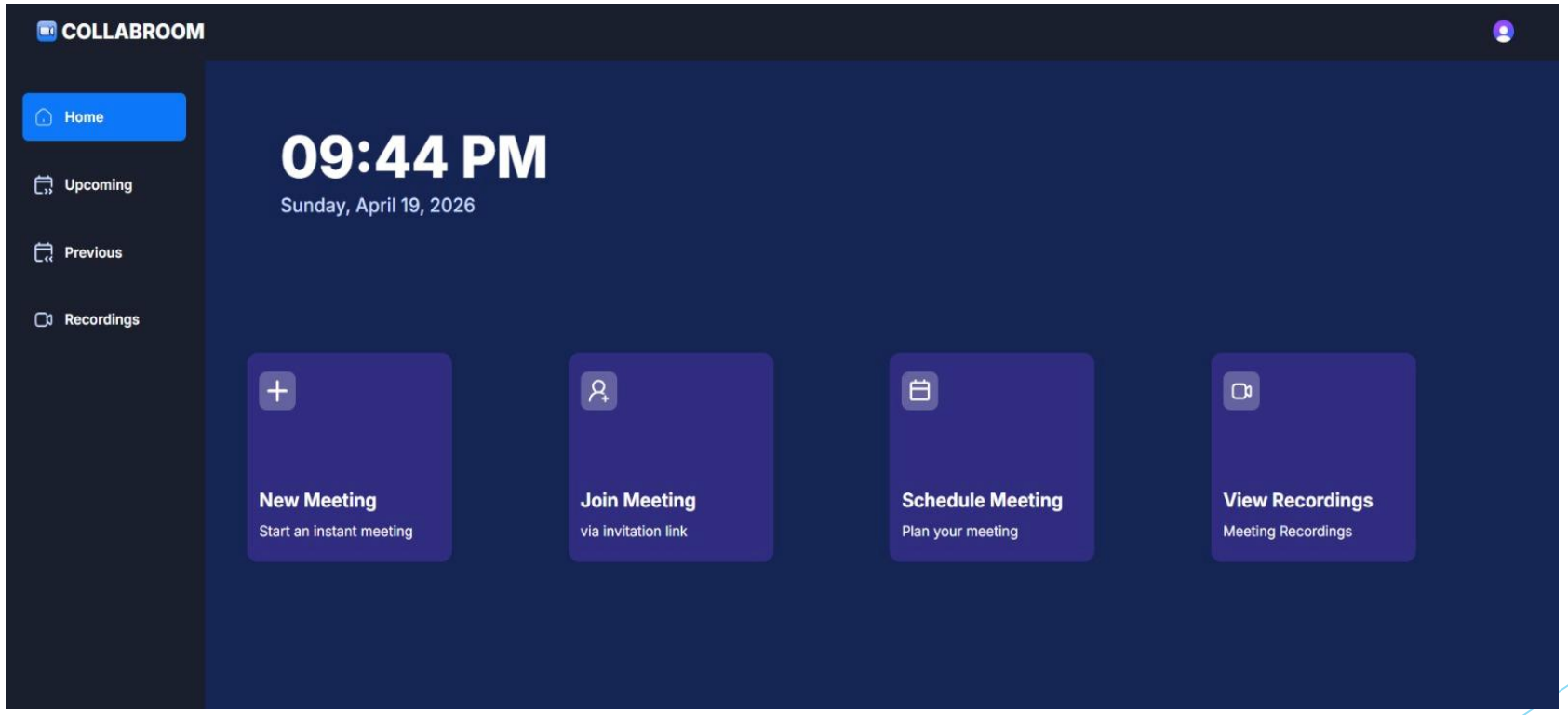
BACKEND
- Node.js
- Stream Video API

TIMELINE OF PROJECT

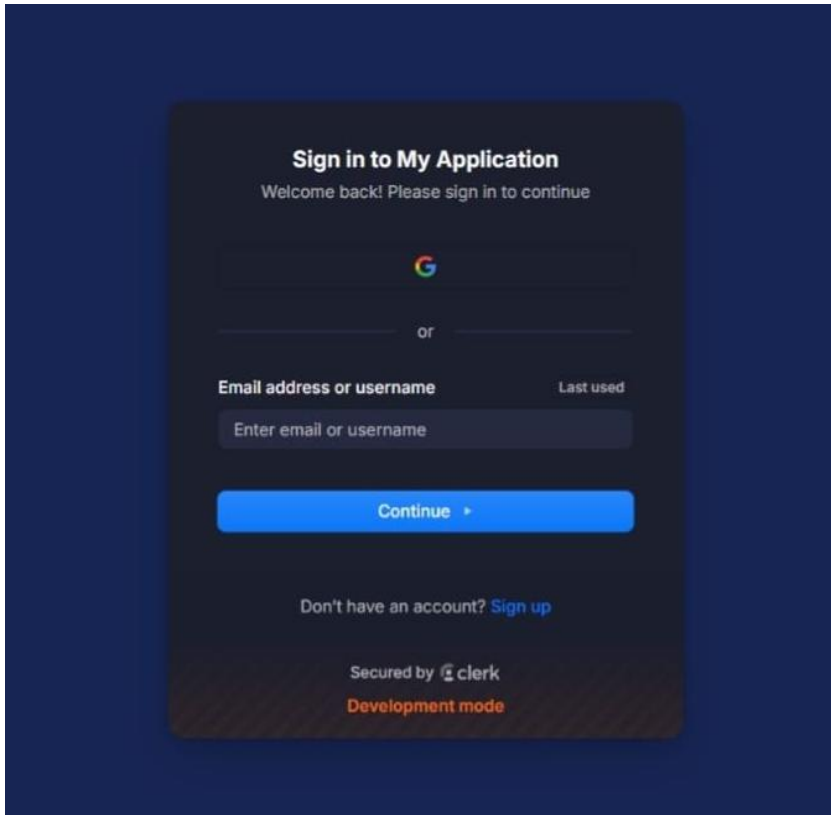
PHASE	DURATION	DESCRIPTION
Requirement Analysis	Week 1	Understanding project goals
Planning and Design	Week 2	UI design and architecture
Development	Week 3-5	Coding frontend and backend
Integration	Week 6	Video API Integration
Testing	Week 7	Debugging and Improvements
Deployment	Week 8	Final deployment
Documentation	Week 9	Report and Presentation

RESULT

1. Home Page

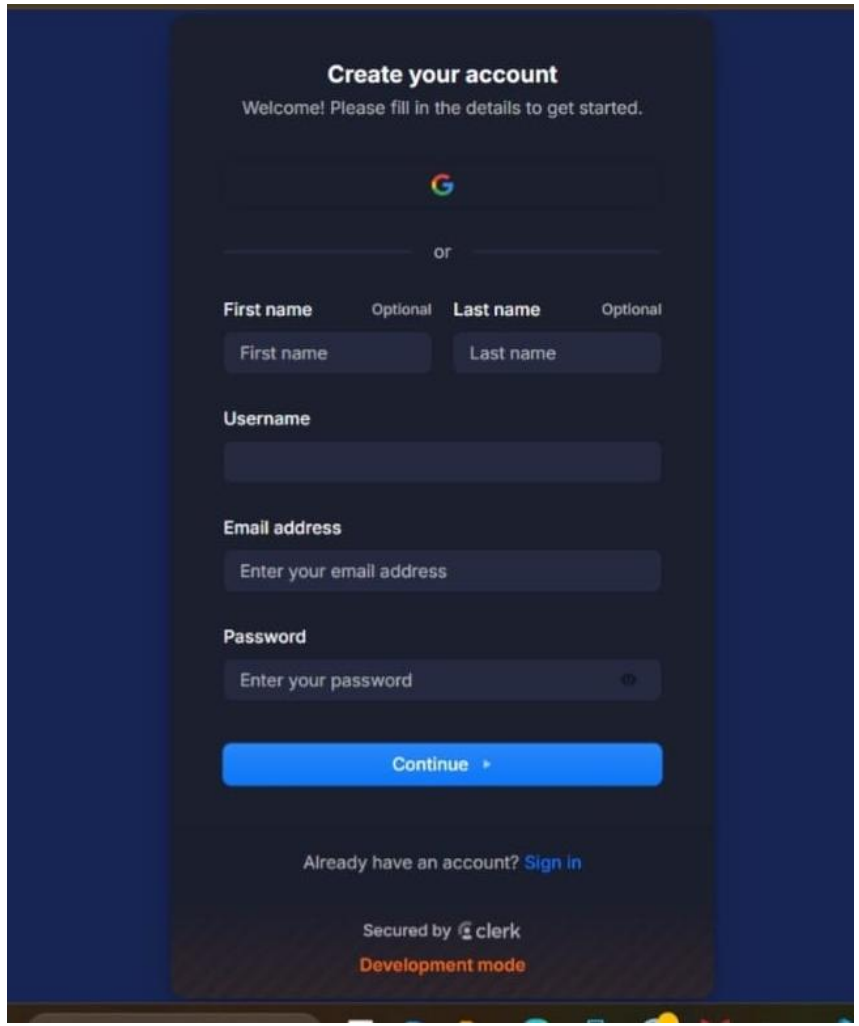


2. Sign in Page



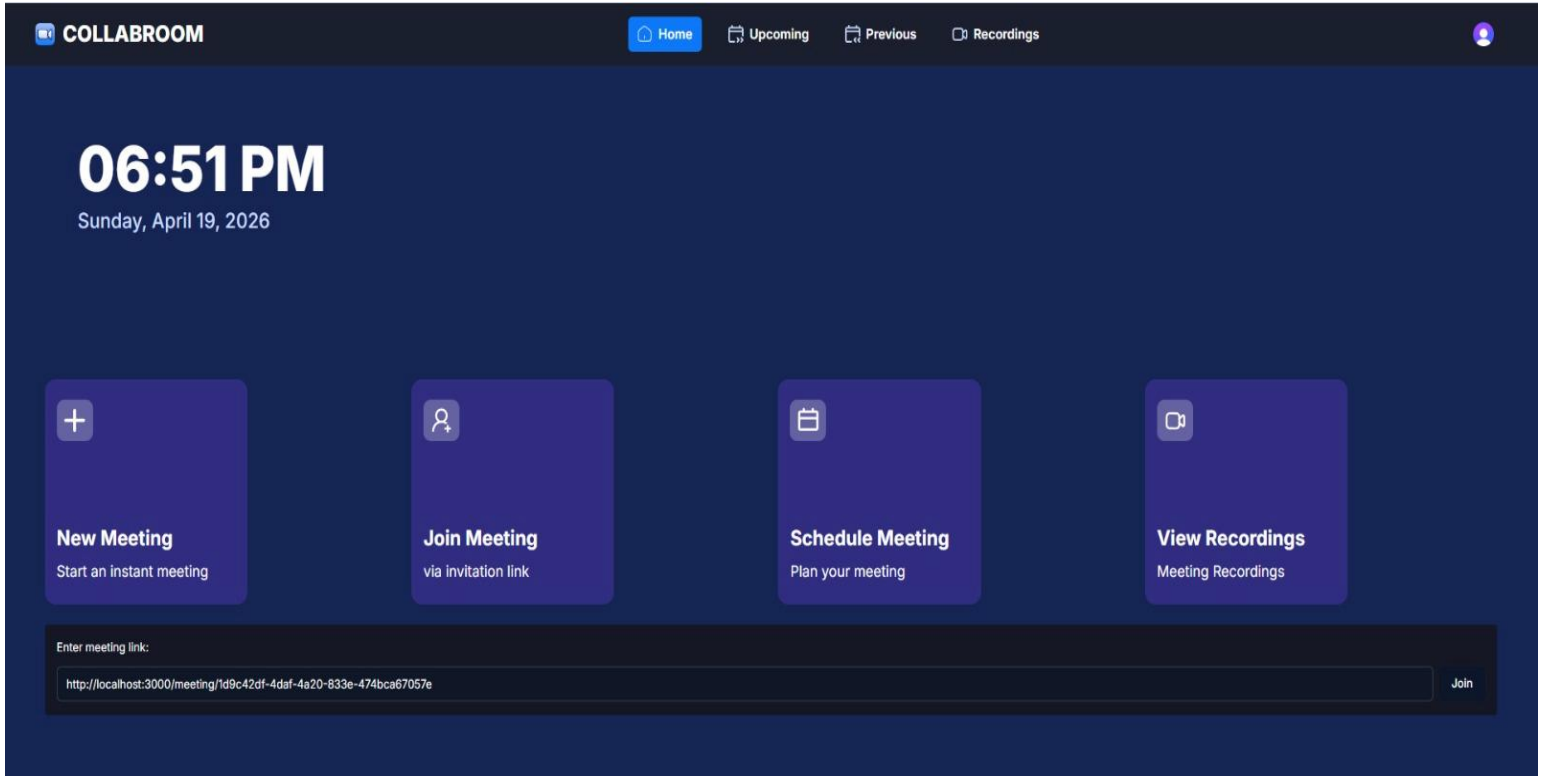
- Enables existing users to log in.
- Authentication using registered email and password.

3. Sign Up Page

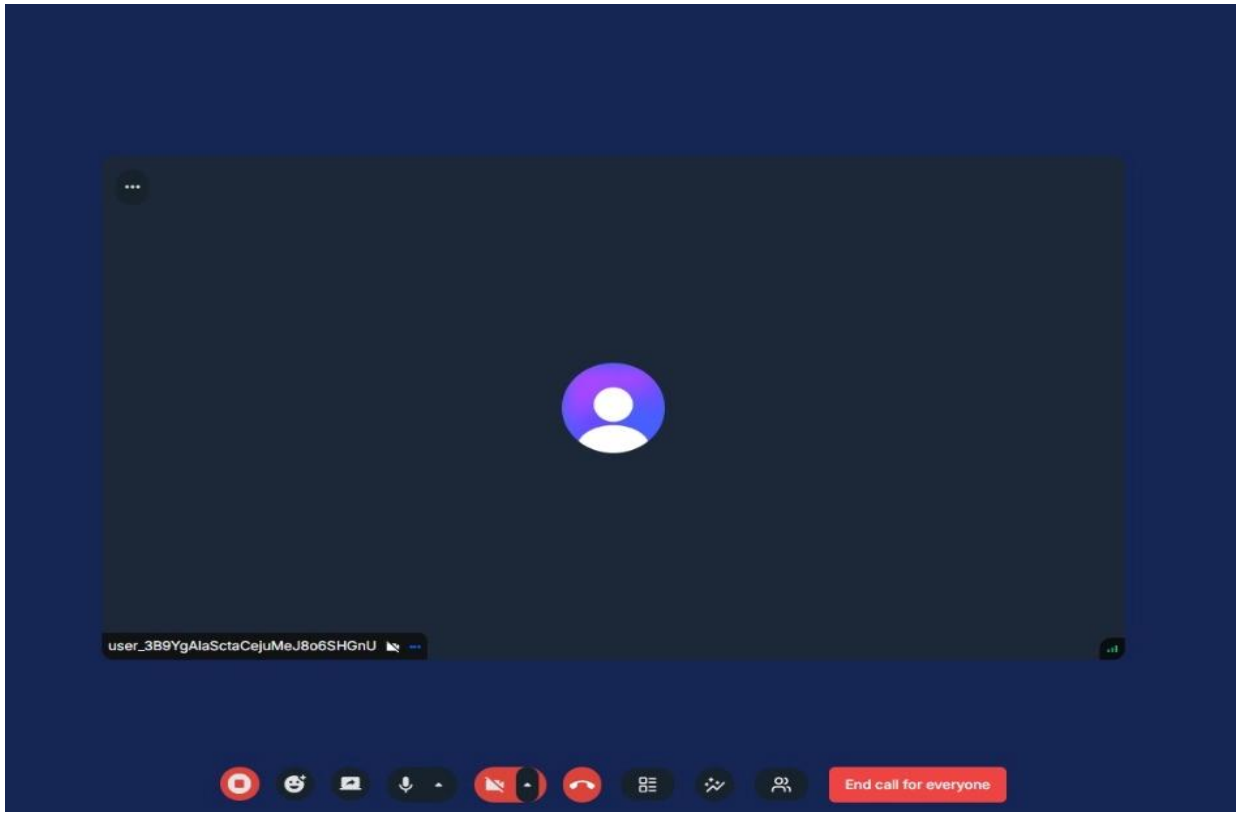


- Allows new users to create an account.
- User enters details such as name, email, and password.
- Data stored in database.

4. New Meeting

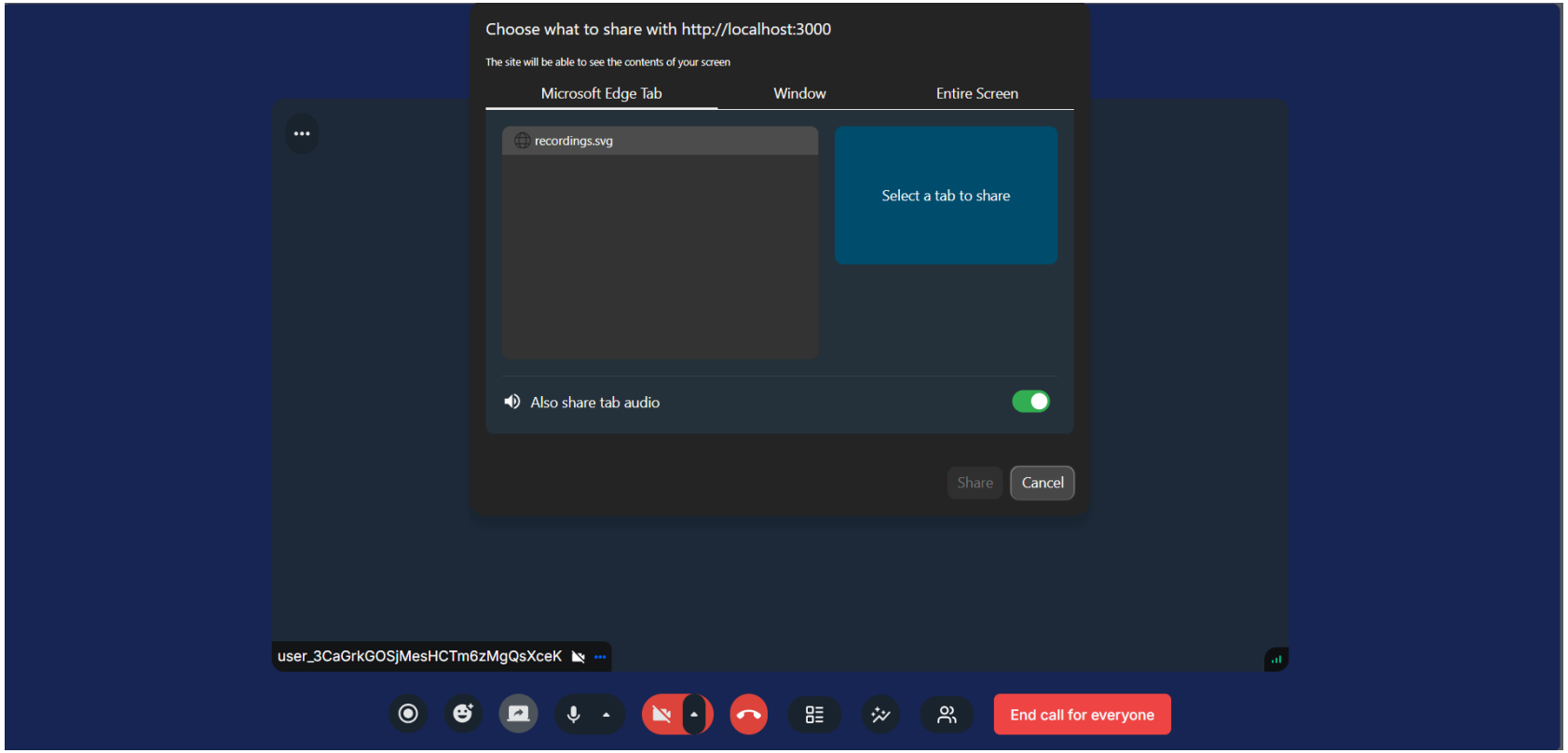


5. Join Meeting

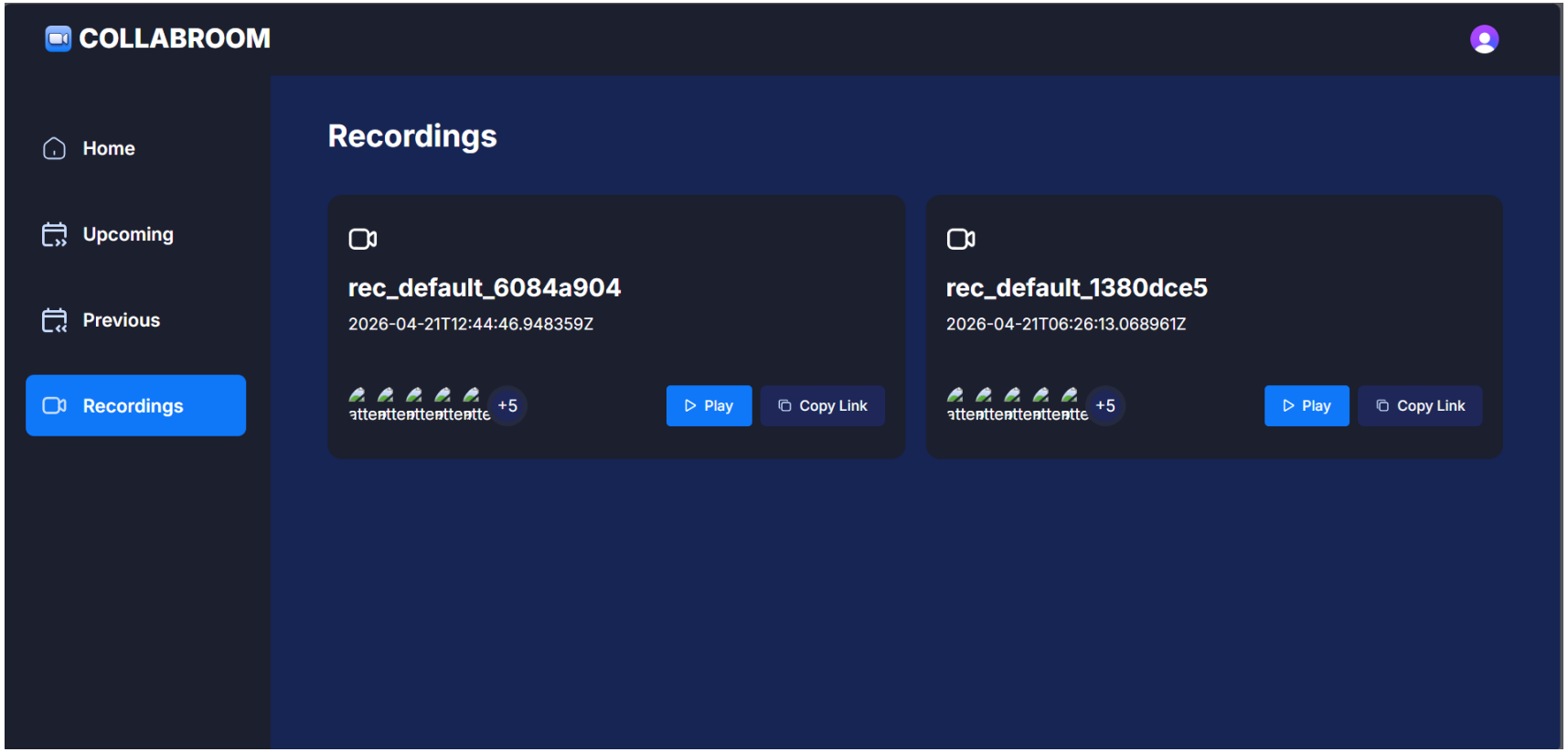


- Users create meetings using Stream API, generating shareable link.
- Clerk authentication ensures only authorized users access meeting link.
- Participants join instantly via link with real-time controls.

6. Screen Sharing



7. Recording


COLLABROOM
Home
Upcoming
Previous
Recordings
Recordings
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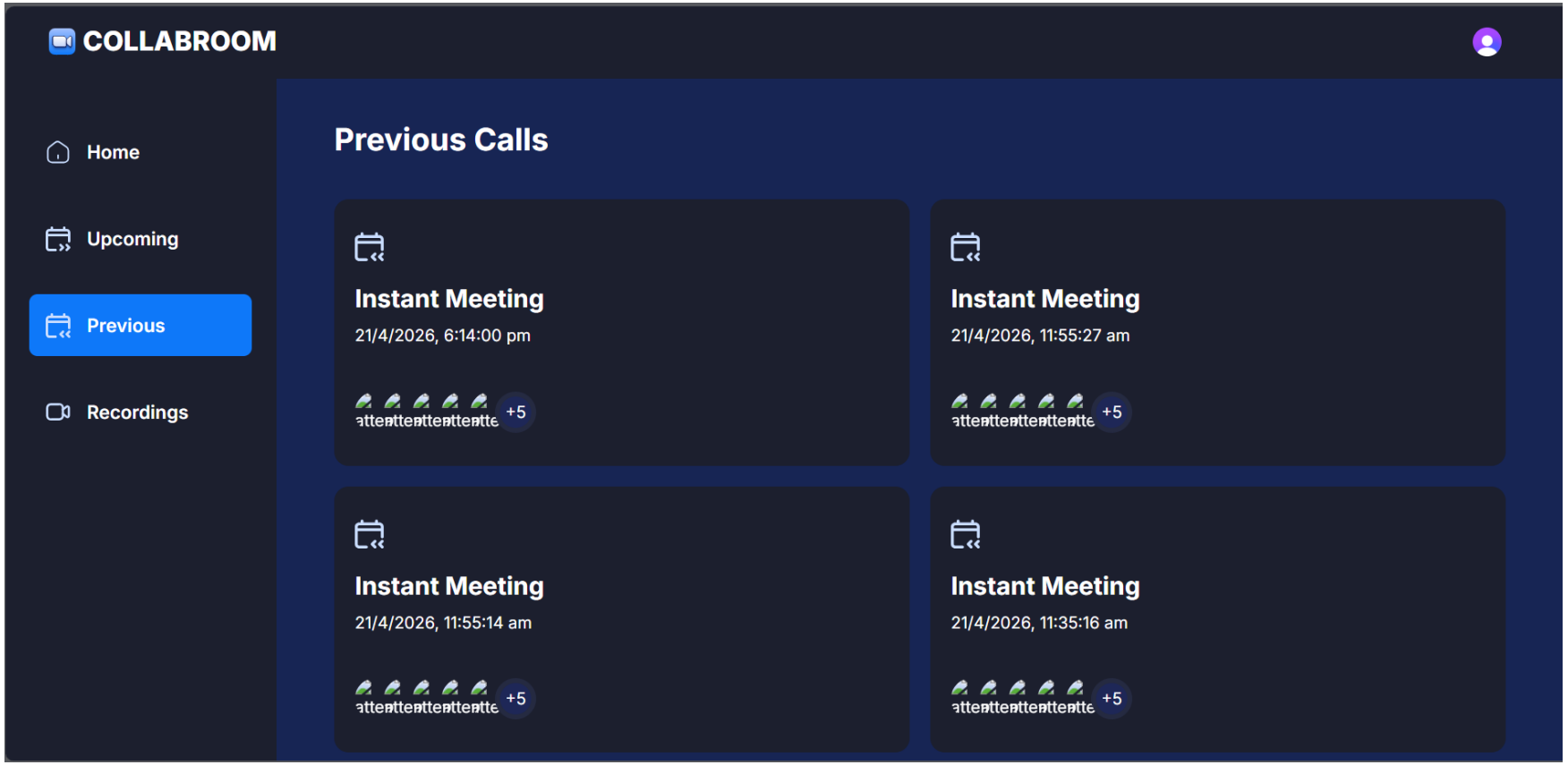
2026-04-21T12:44:46.948359Z
+5
Play
Copy Link
rec_default_1380dce5
2026-04-21T06:26:13.068961Z
+5
Play
Copy Link

8. Previous Meetings



FUTURE ENHANCEMENTS

- **Noise Cancellation:** AI-based background noise cancellation for clearer meeting audio quality.
- **Translation Feature:** Real-time translation and subtitles for seamless multilingual communication experience.
- **Security Enhancement:** Strong end-to-end encryption ensuring secure and private video conferencing sessions.
- **VR Meetings:** VR-based immersive environments for interactive and engaging virtual meetings experience.
- **Smart Assistant** : AI-powered smart assistant for meeting summaries and automated task management.

THANK YOU